ASSESSMENT 4

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QUESTION:

A sports science research centre is investigating variations in exercise performance and pulse rate among fitness-club participants. Using the **Linnerud dataset** available in scikit-learn, analyse chin-ups, sit-ups, jumps, and pulse rate.

Develop two suitable visualisations to examine the distributions and relationships among the selected measurements. Interpret the notable patterns, associations, and variations observed.

(10 Marks)

SOLUTION:

[2]:
```
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.datasets import load_linnerud
```

[4]:
```
data = load_linnerud(as_frame=True)
```

[5]:
```
df = pd.DataFrame({
    'Chin-ups': data.data['Chins'],
    'Sit-ups': data.data['Situps'],
    'Jumps': data.data['Jumps'],
    'Pulse': data.target['Pulse']
})
```

[6]:
```
print("First five observations:")
print(df.head())
```

[7]:
```
print("\nStatistical summary:")
print(df.describe())
```

[12]:
```
df.hist(
    figsize=(10, 7),
    bins=6,
    edgecolor='black'
)
plt.suptitle('Distribution of Exercise Measurements and Pulse')
plt.tight_layout()
plt.show()
```

-

[11]:
```
fig, axes = plt.subplots(1, 3, figsize=(15, 4))

exercise_variables = ['Chin-ups', 'Sit-ups', 'Jumps']

for ax, variable in zip(axes, exercise_variables):
    ax.scatter(df[variable], df['Pulse'])
    ax.set_xlabel(variable)
    ax.set_ylabel('Pulse')
    ax.set_title(variable + ' vs Pulse')
    ax.grid(True, alpha=0.3)

plt.suptitle('Relationship Between Exercise and Pulse')
plt.tight_layout()
plt.show()

# Correlation values for interpretation
print("\nCorrelation with Pulse:")
print(df.corr()['Pulse'].round(2))
```

OUTPUT:

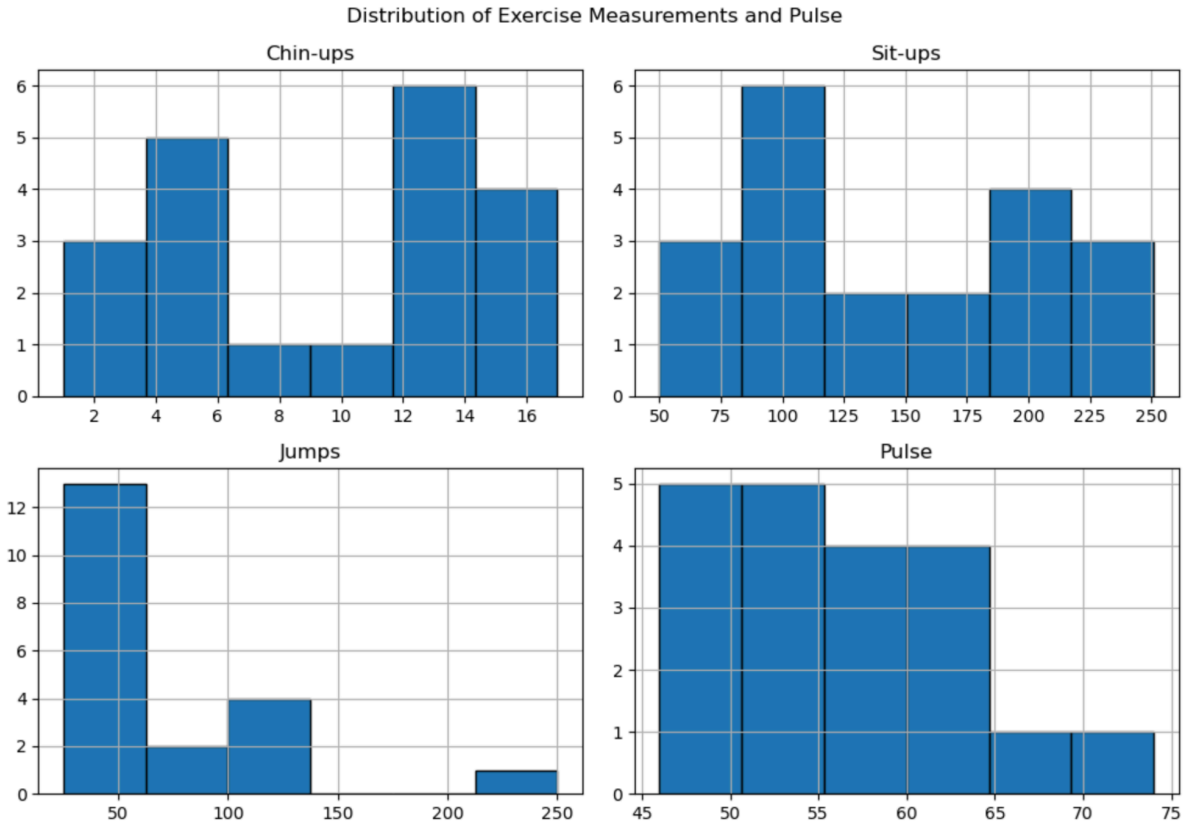
First five observations:

```
First five observations:
   Chin-ups  Sit-ups  Jumps  Pulse
0       5.0    162.0   60.0   50.0
1       2.0    110.0   60.0   52.0
2      12.0    101.0  101.0   58.0
3      12.0    105.0   37.0   62.0
4      13.0    155.0   58.0   46.0
```

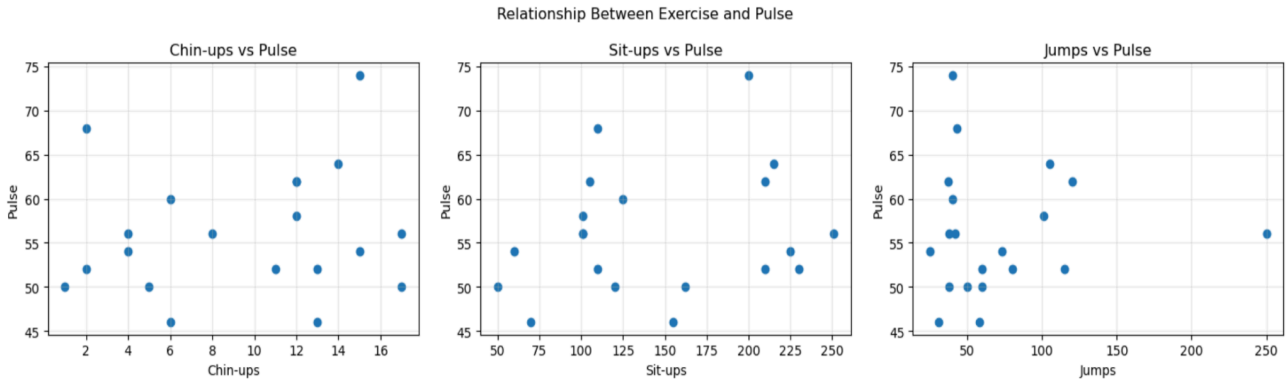
Statistical Summary:

```
Statistical summary:
        Chin-ups     Sit-ups      Jumps      Pulse
count  20.000000   20.000000   20.00000  20.000000
mean    9.450000  145.550000   70.30000  56.100000
std     5.286278   62.566575   51.27747   7.210373
min     1.000000   50.000000   25.00000  46.000000
25%     4.750000  101.000000   39.50000  51.500000
50%    11.500000  122.500000   54.00000  55.000000
75%    13.250000  210.000000   85.25000  60.500000
max    17.000000  251.000000  250.00000  74.000000
```

HISTOGRAM:



SCATTER PLOT:




```
Correlation with Pulse:
Chin-ups    0.15
Sit-ups     0.23
Jumps       0.03
Pulse       1.00
Name: Pulse, dtype: float64
```

INTERPRETATION:

Histogram — Distribution analysis

Chin-ups: Participants show variation in performance, indicating differences in upper-body strength and endurance.
Sit-ups: A wide distribution indicates considerable differences in abdominal endurance among participants.
Jumps: The values show substantial variation, with a notably high observation of 250 that may be an outlier.
Pulse rate: Pulse measurements have a comparatively narrower range than the exercise measurements, suggesting less variation among participants.

Scatter plots — Relationships and associations

Chin-ups vs Pulse: The correlation is approximately 0.15, indicating a very weak positive linear association.
Sit-ups vs Pulse: The correlation is approximately 0.23, indicating a weak positive association.
Jumps vs Pulse: The correlation is approximately 0.03, indicating almost no linear association.